

Mathematics Extension 1

2015 TRIAL HIGHER SCHOOL CERTIFICATE

AM Friday 7th August

Staff Involved:

- VAB* • GIC
- BJB* • WMD
- MRB • DZP
- KJL

90 copies

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen.
Black pen is preferred
- Board-approved calculators may be used.
- Make sure your Barker Student Number is on each page of your solutions.
- On the back of this page is the answer sheet for Section I.
- A table of standard integrals is provided at the back of this paper.
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 2 - 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II Pages 6 – 12

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour 45 minutes for this section.



Barker College

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ANSWER SHEET

Section I – Multiple Choice

Detach this page and hand it in separately.

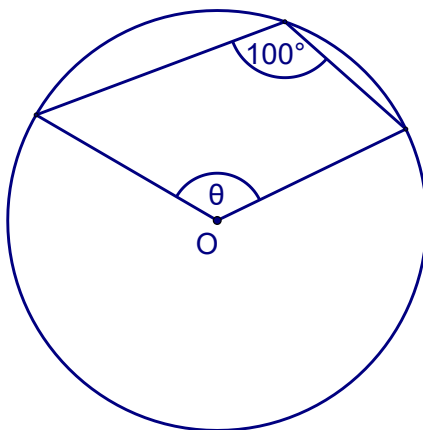
Choose the best response and clearly shade the corresponding circle.

-
- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

2015 Barker College**Trial HSC Extension 1 Mathematics examination****Section I – Multiple Choice (10 marks) Attempt Questions 1–10**

Use the multiple choice answer sheet provided.

1. O is the centre of the circle. Find the value of θ



NOT TO SCALE

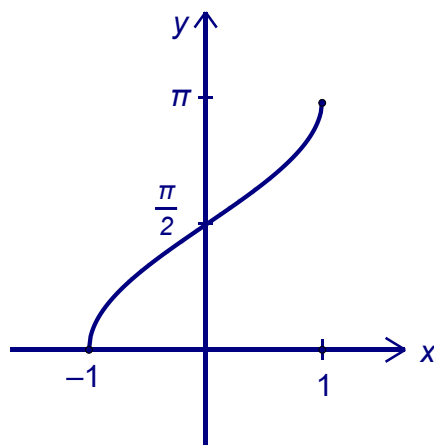
- (A) $\theta = 80^\circ$ (B) $\theta = 160^\circ$
 (C) $\theta = 200^\circ$ (D) $\theta = 260^\circ$
2. $\int \frac{dx}{4x^2 + 16}$ equals which of the following expressions?
- (A) $\frac{1}{8} \tan^{-1} \frac{x}{2} + C$ (B) $\frac{1}{4} \tan^{-1} \frac{x}{2} + C$
 (C) $\frac{1}{2} \tan^{-1} (2x) + C$ (D) $\frac{1}{4} \tan^{-1} (2x) + C$
3. Choose the expression which is equivalent to $\sqrt{\cos^2 4\theta - \sin^2 4\theta}$
- (A) $\cos 2\theta - \sin 2\theta$ (B) $\cos 4\theta - \sin 4\theta$
 (C) $\sqrt{\cos 8\theta}$ (D) $\sqrt{1 - 2\sin^2 2\theta}$

Section I – Multiple Choice (continued)

4. Choose the expression that is equivalent to $3 \sin x + \sqrt{3} \cos x$

- (A) $2\sqrt{3} \sin\left(x + \frac{\pi}{3}\right)$ (B) $2\sqrt{3} \sin\left(x + \frac{\pi}{6}\right)$
 (C) $2\sqrt{3} \sin\left(x - \frac{\pi}{3}\right)$ (D) $2\sqrt{3} \sin\left(x - \frac{\pi}{6}\right)$

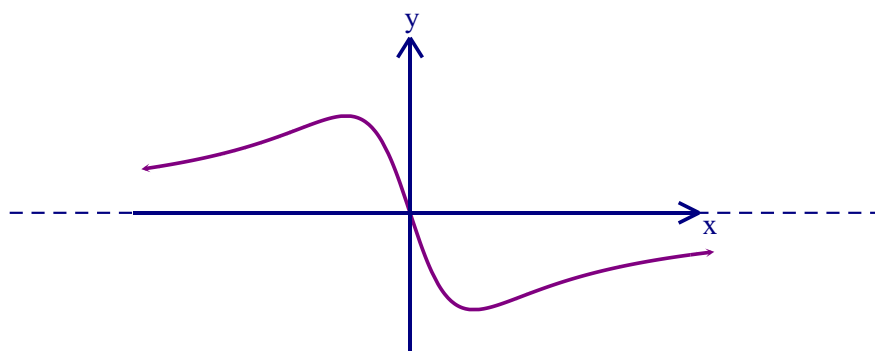
5.



Which combination of the two equations $y = -\cos^{-1} x$ and $y = \frac{\pi}{2} + \sin^{-1} x$ describe the curve shown above?

- (A) Both the equations (B) The first equation only
 (C) The second equation only (D) Neither of the equations

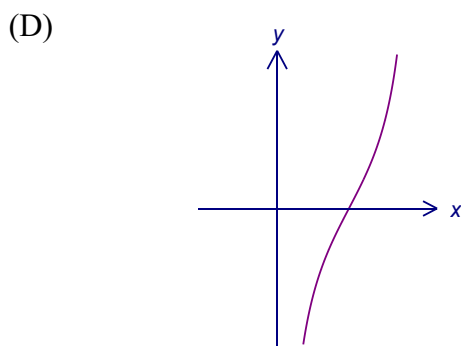
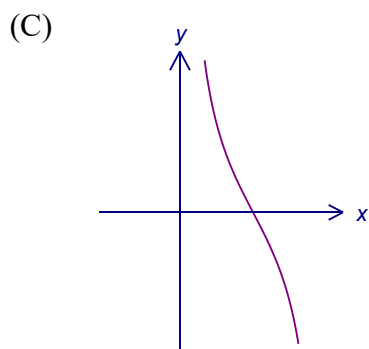
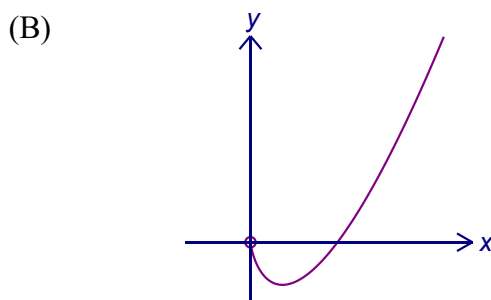
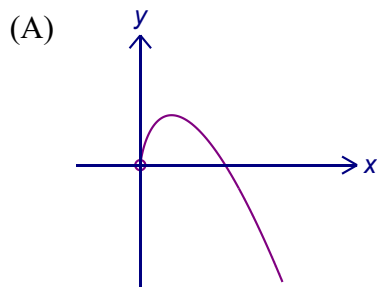
6. One of the four equations below describes this graph. Choose the correct equation.



- (A) $y = \frac{1}{1+x^2}$ (B) $y = 1 - \frac{1}{1+x^2}$
 (C) $y = \frac{x}{1+x^2}$ (D) $y = \frac{-x}{1+x^2}$

Section I – Multiple Choice (continued)

7. Select the graph which could represent the function $y = x \log_e x$



8. Three of these statements are always true and one is sometimes false.
Select the statement which is sometimes false:

(A) $\int f(x)dx = \int f(y)dy$ (B) $\int_a^b f(x)dx = -\int_b^a f(x)dx$

(C) $x - \int f(x)dx = \int (1 - f(x))dx$ (D) $\int_{-a}^a f(x)dx = 2\int_0^a f(x)dx$

9. What is the general solution to the equation $\sin 2x = 1$, where n is any integer?

(A) $x = \frac{\pm\pi}{4} + n\pi$ (B) $x = \frac{\pm\pi}{4} + 2n\pi$

(C) $x = \frac{\pi}{4} + n\pi$ (D) $x = \frac{\pi}{4} + 2n\pi$

10. For the function $y = \sin^{-1}(1 - x^2)$ what is the natural domain?

(A) $-\sqrt{2} \leq x \leq \sqrt{2}$ (B) $-1 \leq x \leq 1$

(C) $0 \leq x \leq \sqrt{2}$ (D) $0 \leq x \leq 1$

End of Section 1

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

Extra writing booklets are available.

In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Question 11 (15 marks) [Use a SEPARATE writing booklet]

Marks

(a) Solve $\frac{x^2 + 20}{x - 4} < -4$ **3**

(b) Find

(i) $\int \frac{1}{\sqrt{\frac{1}{9} - x^2}} dx$ **1**

(ii) $\int \sin^2 x dx$ **2**

Question 11 continues on page 7

- (c) A population of birds on an island changes according to the equation $\frac{dP}{dt} = k(P - 3500)$. Initially there were 40 individuals and after 5 years the population had increased to 125.

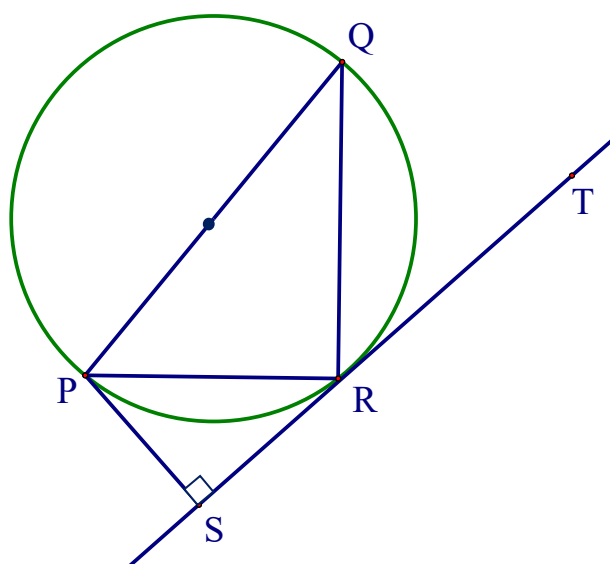
- (i) Verify that $P = 3500 - Ae^{kt}$ satisfies the above equation, where P is the population, t is the time in years and A and k are constants. 1
- (ii) Find the value of A and find the value of k to 3 significant figures. 3
- (iii) Sketch the function $P = 3500 - Ae^{kt}$ for $t \geq 0$ showing any intercepts and asymptotes. 2

- (d) In the diagram PQ is a diameter of the circle. TS is a tangent to the circle at R and PS is the interval from P perpendicular to TS.

Copy or trace the diagram into your examination booklet.

Prove that PR bisects $\angle QPS$

3



NOT TO SCALE

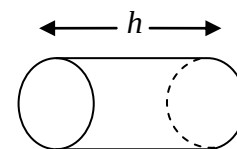
End of Question 11

- (a) A sculptor is rolling modelling clay on a board.

The clay is in the shape of a cylinder and has constant volume $20\pi \text{ cm}^3$.

As she rolls it into a thinner cylinder, the radius decreases at a constant rate of 2 cm per minute.

Find the rate of increase of the height h of the cylinder with respect to time, at the moment when the radius is 4 cm.



3

- (b) Prove by mathematical induction that for all positive integers n

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n \times (n+1)!$$

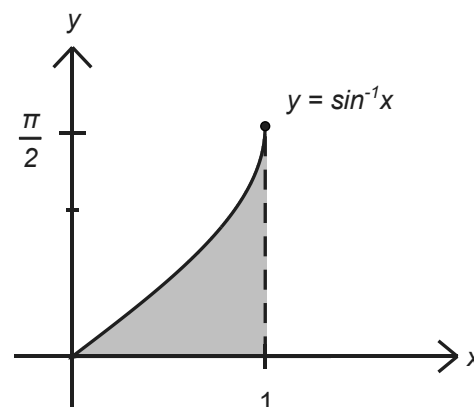
4

- (c) (i) Show that the derivative of $x \sin^{-1} x + \sqrt{1-x^2}$ is $\sin^{-1} x$

2

- (ii) Hence, determine the shaded area shown which is bounded by the curve $y = \sin^{-1} x$, the x -axis and the line $x = 1$

2



- (d) The function $f(x) = \frac{1}{2}(x-4)^2$ is shown.

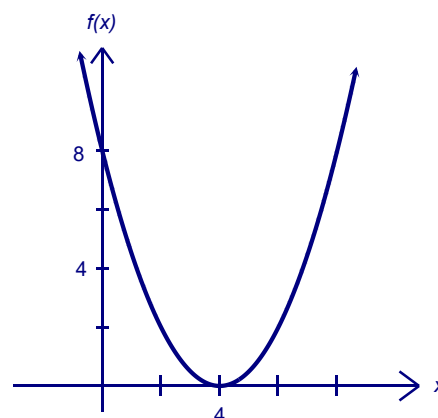
$g(x) = f(x)$ over the limited domain $x \geq 4$

- (i) Find $g^{-1}(x)$, the inverse function of $g(x)$

2

- (ii) Find the point of intersection of $g(x)$ and $g^{-1}(x)$

2



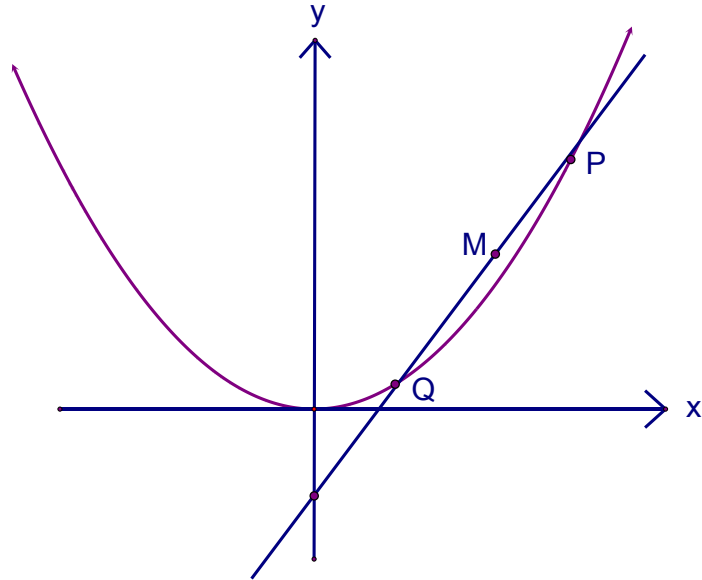
End of Question 12

- (a) A class consists of 12 girls and 10 boys.

How many selections of 4 students are there which contain at least 2 girls?

2

- (b)



$P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$

- (i) Write down the gradient of the chord PQ .

1

- (ii) Hence show the equation of the chord PQ is $y - \frac{1}{2}(p+q)x + apq = 0$

2

- (iii) P and Q move so that the chord PQ passes through the point $(0, -a)$.

Show that $pq = 1$.

1

- (iv) Given that M is the midpoint of PQ with coordinates

$$\left[a(p+q), \frac{1}{2}a(p^2 + q^2) \right] \text{ (DO NOT PROVE THIS),}$$

find the locus of M .

2

Question 13 continues on page 10

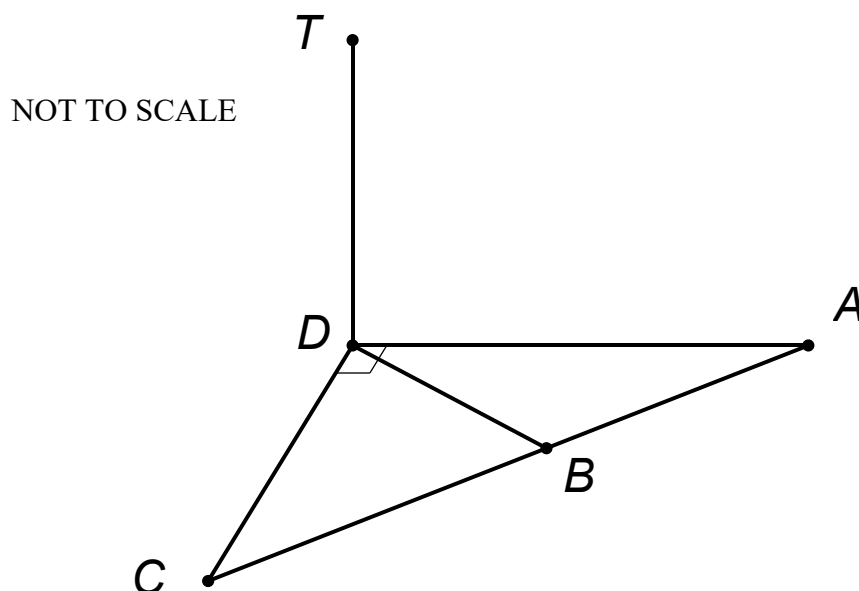
- (c) A particle moves according to the equation $v^2 = 16 - 4x^2$ where x is its position in metres and v is its velocity in metres per second.

Initially the particle is at the origin and moves off in a positive direction.

- | | | |
|-------|---|----------|
| (i) | Prove the particle has acceleration in the form $a = -n^2x$ where n is a constant. | 1 |
| (ii) | Find the amplitude of the motion, explaining how this is found. | 2 |
| (iii) | Use integration to find an expression for x as a function of time t , beginning with $v = \sqrt{16 - 4x^2}$ | 4 |

End of Question 13

- (a) Three surveyors A , B and C are observing a tower of height h situated on flat ground. A is due east of the tower, C is due south of it and B is somewhere between A and C on the line AC . The tower's base is at D and its top is at T .
- The angles of elevation of the top of the tower from A , B and C are 17° , 20° and 24° respectively. Note that $\angle ABD$ is obtuse.



- | | | |
|-------|---|----------|
| (i) | Show $AD = h \tan 73^\circ$ and write similar expressions for BD and CD . | 1 |
| (ii) | Find $\angle DAC$ to the nearest minute. | 1 |
| (iii) | Hence, find the bearing of B from the base of the tower D . | 3 |

Question 14 continues on page 12

Question 14 (continued)

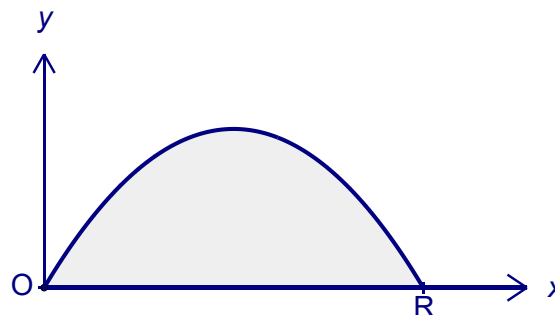
Marks

- (b) A projectile is fired at V m/s at an angle θ with the horizontal.

The equations of motion of the particle are

$$x = Vt \cos \theta, \quad y = -\frac{1}{2}gt^2 + Vt \sin \theta$$

DO NOT PROVE THIS



- (i) Using the above equations, show the Cartesian equation of motion of the particle

$$\text{is } y = \frac{-g}{2V^2 \cos^2 \theta} x^2 + x \tan \theta$$

1

- (ii) The range of the projectile is given by

$$R = \frac{2V^2 \sin \theta \cos \theta}{g} \quad (\text{DO NOT PROVE THIS})$$

Show that the shaded area A under the path of the parabola, for fixed values of

$$V \text{ and } \theta, \text{ is } A = \frac{2V^4}{3g^2} \sin^3 \theta \cos \theta$$

3

- (iii) The shaded area shown is to be illuminated by rockets fired from O , fitted with downward-directed lights.

If θ is allowed to vary while V is kept constant, there is a maximum value of A for a value of θ between 0 and $\frac{\pi}{2}$.

Find this value of θ .

2

- (c) (i) Prove that $\cos h - 1 = -2 \sin^2 \frac{h}{2}$

1

- (ii) Hence, given that $f(x) = \cos x$, find $f'(x)$ from first principles,

$$\text{using the definition } f'(x) = \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right]$$

3

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Trial HSC Ext 1 SOLUTIONS 2015

1 Reflex angle at $\theta = 200^\circ$

$$\therefore \theta = 360^\circ - 200^\circ = 160^\circ$$

B

$$\frac{2}{\int \frac{1}{4x^2+16} dx = \frac{1}{4} \int \frac{1}{x^2+4} dx}$$

$$= \frac{1}{4} \times \frac{1}{2} \tan^{-1} \frac{x}{2} + c$$

A

$$\frac{3}{\cos^2 x - \sin^2 x = \cos 2x}$$

$$\therefore \sqrt{\cos^2 4\theta - \sin^2 4\theta}$$

$$= \sqrt{\cos 8\theta} \quad C$$

$$\frac{4}{\text{Let } 3\sin x + \sqrt{3}\cos x}$$

B

$$\equiv R \sin(x+\alpha)$$

$$\equiv R \sin x \cos \alpha + R \cos x \sin \alpha$$

Equate coefficients

$$\therefore 3 = R \cos \alpha, \quad \sqrt{3} = R \sin \alpha \quad \text{--- (1)}$$

$$\frac{(2)}{(1)} \therefore \frac{R \sin \alpha}{R \cos \alpha} = \frac{\sqrt{3}}{3}$$

$$\therefore \tan \alpha = \frac{1}{\sqrt{3}}, \quad \therefore \alpha = \frac{\pi}{6}$$

$$(1)^2: 9 = R^2 \cos^2 \alpha \quad \text{--- (3)}$$

$$(2)^2: 3 = R^2 \sin^2 \alpha \quad \text{--- (4)}$$

$$(3) + (4) \quad 12 = R^2 (\cos^2 \alpha + \sin^2 \alpha)$$

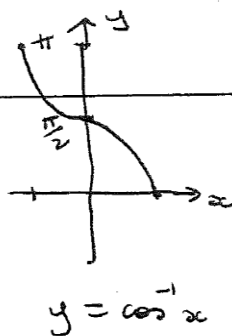
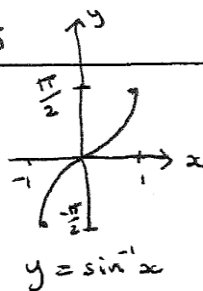
$$\therefore 12 = R^2$$

$$\therefore R = \sqrt{12} = 2\sqrt{3}$$

$$\therefore 3\sin x + \sqrt{3}\cos x$$

$$= 2\sqrt{3} \sin(x + \frac{\pi}{6})$$

5
C
2nd equation only



shift up $\frac{\pi}{2}$ to

$$\text{get } y = \frac{\pi}{2} + \sin^{-1} x$$

Yes, matches graph.

But $y = -\cos^{-1} x$ is reflected in x-axis, not y-axis
No, does not match graph.

6 A does not pass through $(0,0)$

B is even, not odd x

C is in Quadrants 1 and 3 x

D ✓

$$\frac{7}{B \text{ for } 0 < x < 1, \quad y < 0 \quad \left. \begin{array}{l} \text{for } x > 1, \\ y > 0 \end{array} \right\}$$

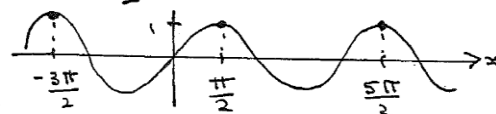
and as $x \rightarrow 0, y \rightarrow 0$

8 D

D is only true for even functions and is false for other functions

$$\frac{9}{\sin 2x = 1} \quad C$$

$$\therefore 2x = \frac{\pi}{2} + 2n\pi$$



$$\therefore x = \frac{\pi}{4} + n\pi$$

$$\frac{10}{y = \sin^{-1}(1-x^2)} \quad A$$

Domain of $\sin^{-1} m$ is $-1 \leq m \leq 1$

$$\therefore -1 \leq 1-x^2 \leq 1$$

$$\therefore -2 \leq -x^2 \leq 0$$

$$\therefore 0 \leq x^2 \leq 2$$

$$\therefore -\sqrt{2} \leq x \leq \sqrt{2}$$

- | | |
|------|-------|
| 1. B | 6. D |
| 2. A | 7. B |
| 3. C | 8. D |
| 4. B | 9. C |
| 5. C | 10. A |

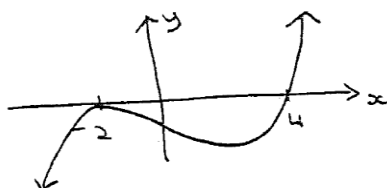
11 a) $\frac{x^2+20}{x-4} < -4$

$\therefore \frac{x^2+20}{x-4} + 4 < 0$

$\therefore \frac{x^2+20+4x-16}{x-4} < 0$

$\therefore \frac{x^2+4x+4}{x-4} < 0$

$\therefore (x+2)^2(x-4) < 0$



$\therefore x < -2, -2 < x < 4$

b) i) $\int \frac{1}{\sqrt{\frac{1}{9} - x^2}} dx = \sin^{-1} \frac{x}{\frac{1}{3}} + C$
 $= \sin^{-1} 3x + C$

b) ii) $\int \sin^2 x dx = \int \frac{1}{2}(1 - \cos 2x) dx$
 $= \frac{1}{2}(x - \frac{1}{2}\sin 2x) + C$
 $= \frac{1}{2}x - \frac{1}{4}\sin 2x + C$

c) i) For $P = 3500 - Ae^{kt}$

$\therefore \frac{dP}{dt} = -Ake^{kt}$

and $k(P - 3500)$

$= k(3500 - Ae^{kt} - 3500)$

$= -kAe^{kt}$

$\therefore \frac{dP}{dt} = k(P - 3500)$ as required
 $\therefore$ it is a solution

11 c) ii) $\left. \begin{matrix} t=0 \\ P=40 \end{matrix} \right\} \Rightarrow P = 3500 - Ae^{kt}$

$\therefore 40 = 3500 - A$

$\therefore A = 3460$

$\left. \begin{matrix} t=5 \\ P=125 \end{matrix} \right\} \Rightarrow P = 3500 - 3460e^{5k}$
 $\therefore 125 = 3500 - 3460e^{5k}$

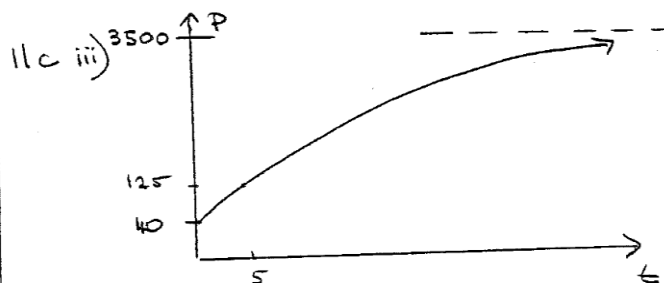
$\therefore 3460e^{5k} = 3375$

$\therefore e^{5k} = \frac{3375}{3460} = 0.97543 \dots$

$\therefore 5k = \log_e 0.97543 \dots$

$\therefore k = \frac{1}{5} \log_e 0.97543 \dots$

$\therefore k = -0.00497$ (3 s.f.)



d)

$\angle PRQ = 90^\circ$ ($\angle$ in a semicircle)

$\angle PRS = \angle PQR$ ($\angle$ between chord and tangent = $\angle$ in alternate segment)

$\angle RPS = 90^\circ - \angle PRS$ ($\angle$ sum Δ RPS)

$\angle QPR = 180^\circ - \angle PRQ - \angle PQR$
 ($\angle$ sum Δ PQR)

$= 180^\circ - 90^\circ - \angle PRS$ (proved above)

$= 90^\circ - \angle PRS$

$\therefore \angle RPS = \angle QPR$ (both equal $90^\circ - \angle PRS$)

$\therefore RP$ bisects $\angle QPS$

$$12a) 20\pi = \pi r^2 h$$

$$\therefore 20 = r^2 h$$

$$\therefore h = 20r^{-2}$$

$$\therefore \frac{dh}{dr} = -40r^{-3} = \frac{-40}{r^3}$$

$$\text{Find } \frac{dh}{dt} = ?$$

$$\frac{dh}{dt} = \frac{dh}{dr} \times \frac{dr}{dt}$$

$$= \frac{-40}{r^3} \times -2$$

$$= \frac{80}{r^3}$$

$$\therefore \text{When } r = 4$$

$$\frac{dh}{dt} = \frac{80}{4^3} = 1.25 \text{ cm/min}$$

$$12c) i) \text{ Let } y = x \sin^{-1} x + \sqrt{1-x^2}$$

$$\therefore y = x \sin^{-1} x + (1-x^2)^{\frac{1}{2}}$$

$$\therefore \frac{dy}{dx} = \sin^{-1} x + \frac{x}{\sqrt{1-x^2}} + \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \times -2x$$

$$= \sin^{-1} x + \frac{x}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}}$$

$$= \sin^{-1} x$$

$$c) ii) \text{ Area} = \int_0^1 \sin^{-1} x \, dx$$

$$= \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^1$$

$$= \left(\sin^{-1} 1 + 0 - 0 - 1 \right)$$

$$= \left(\frac{\pi}{2} - 1 \right) \text{ unit}^2$$

$$12b) \text{ When } n=1, \text{ LHS} = 2 \times 1! = 2, \text{ RHS} = 1 \times 2! = 2 = \text{LHS} \therefore \text{true for } n=1$$

Assume true some integer $n=k$

$$\therefore \text{Assume } 2 \times 1! + 5 \times 2! + \dots + (k^2+1)k! = k(k+1)!$$

$$\text{Aim: Prove } 2 \times 1! + 5 \times 2! + \dots + (k^2+1)k! + [(k+1)^2+1](k+1)! = (k+1)(k+2)!$$

$$\text{Proof } 2 \times 1! + 5 \times 2! + \dots + (k^2+1)k! + [(k+1)^2+1](k+1)!$$

$$= k(k+1)! + (k^2+2k+2)(k+1)! \quad (\text{by assumption})$$

$$= (k+1)! (k^2+3k+2)$$

$$= (k+1)! (k+2)(k+1) = (k+2)! (k+1)$$

$\therefore$ If true for $n=k$
proved true for $n=k+1$
 $\therefore$ By math ind'n true
for all positive integers n

12d i)

$$\therefore f(x) \text{ is } y = \frac{1}{2}(x-4)^2$$

$$\therefore f^{-1}(x) \text{ is } x = \frac{1}{2}(y-4)^2$$

$$\therefore 2x = (y-4)^2$$

$$\therefore \pm \sqrt{2x} = y-4$$

$$\therefore y = 4 \pm \sqrt{2x}$$

$$\therefore y = 4 + \sqrt{2x}$$

Since domain
of $f(x)$ is $x \geq 4$
so range of $f^{-1}(x)$
is $y \geq 4$

$$12d ii) f(x) \text{ and } f^{-1}(x)$$

intersect on line $y=x$

$$\therefore \text{Solve } f(x)=x$$

$$\therefore \frac{1}{2}(x-4)^2 = x$$

$$\therefore x^2 - 8x + 16 = 2x$$

$$\therefore x^2 - 10x + 16 = 0$$

$$\therefore (x-8)(x-2) = 0$$

$$\therefore x = 2 \text{ or } 8$$

$$\therefore P_1 = (2, 2) \text{ or } (8, 8)$$

But $y > 4$

$$\therefore P_1 \text{ is } (8, 8)$$

13a) Selections are 2g2b or 3g1b or 4g

$$\begin{aligned}\text{No selections} &= {}^{12}C_2 \cdot {}^{10}C_2 + {}^{12}C_3 \cdot {}^{10}C_1 + {}^{12}C_4 \\ &= 2970 + 2200 + 495 \\ &= 5665\end{aligned}$$

13b) i) $m = \frac{p+q}{2}$

ii) $y - ap^2 = \frac{p+q}{2}(x - 2ap)$

$$\therefore y - ap^2 = \frac{1}{2}(p+q)x - ap(p+q)$$

$$\therefore y - ap^2 - \frac{1}{2}(p+q)x + ap(p+q) = 0$$

$$\therefore y - ap^2 - \frac{1}{2}(p+q)x + ap^2 + apq = 0$$

$$\therefore y - \frac{1}{2}(p+q)x + apq = 0$$

iii) Sub $(0, -a)$ in (ii)

$$\therefore -a + apq = 0$$

$$\therefore apq = a$$

$$\therefore pq = 1$$

iv) P + M

$$\begin{aligned}X &= a(p+q) & Y &= \frac{1}{2}a(p^2+q^2) \\ \therefore \frac{X}{a} &= p+q \text{ --- (1)} & \therefore 2Y &= a(p^2+2pq+q^2-2pq) \\ & & &= a(p^2+2pq+q^2-2) \\ & & \therefore 2Y &= a(p+q)^2 - 2a \text{ --- (2)}\end{aligned}$$

Sub (1) in (2) $\therefore 2Y = a \times \frac{X^2}{a^2} - 2a$

$$\therefore 2Y = \frac{X^2}{a} - 2a$$

$$\therefore 2aY = X^2 - 2a^2$$

$$\therefore X^2 = 2a^2 + 2aY$$

or $X^2 = 2a(Y+a)$

$$13c) i) v^2 = 16 - 4x^2$$

$$a = \frac{dv}{dx} \left(\frac{1}{2} v^2 \right) \\ = \frac{d}{dx} (8 - 2x^2) \\ = -4x \\ = -2^2 x \\ \text{as required}$$

$$c) ii) \text{ When } v = 0, 16 - 4x^2 = 0 \\ \therefore x^2 = 4 \\ \therefore x = \pm 2$$

These are the two extreme positions of motion.

$\therefore$ Amplitude = 2 metres.

$$c) iii) v = \sqrt{16 - 4x^2}$$

$$\therefore \frac{dx}{dt} = \sqrt{16 - 4x^2}$$

$$\therefore \frac{dt}{dx} = \frac{1}{\sqrt{16 - 4x^2}} =$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{4 - x^2}}$$

$$\therefore t = \frac{1}{2} \sin^{-1} \left(\frac{x}{2} \right) + C$$

$$\text{When } t = 0 \left. \begin{array}{l} x = 0 \end{array} \right\} \therefore 0 = 0 + C \\ \therefore C = 0$$

$$\therefore t = \frac{1}{2} \sin^{-1} \left(\frac{x}{2} \right)$$

$$\therefore 2t = \sin^{-1} \left(\frac{x}{2} \right)$$

$$\therefore \frac{x}{2} = \sin(2t)$$

$$\therefore x = 2 \sin(2t)$$

$$14a) i) \text{ In } \triangle ADT \quad \tan 73^\circ = \frac{AD}{h}$$

$$\therefore AD = h \tan 73^\circ$$

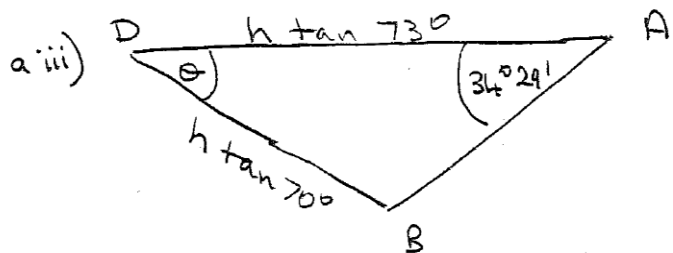
$$BD = h \tan 70^\circ$$

$$CD = h \tan 66^\circ$$

$$a) ii) \text{ In } \triangle ACD$$

$$\tan \angle DAC = \frac{CD}{AD} = \frac{h \tan 66^\circ}{h \tan 73^\circ} \\ = 0.68668 \dots$$

$$\therefore \angle DAC = 34^\circ 29'$$



$$\frac{\sin B}{h \tan 73^\circ} = \frac{\sin 34^\circ 29'}{h \tan 70^\circ}$$

$$\therefore \sin B = \frac{h \tan 73^\circ \sin 34^\circ 29'}{h \tan 70^\circ} \\ = 0.674 \dots$$

$$\therefore \angle B = \angle ABD = 42^\circ 23' \text{ or } 137^\circ 37'$$

But $\angle ABD$ is obtuse (data)

$$\therefore \angle ABD = 137^\circ 37'$$

$$\therefore \angle ADB = 180^\circ - 137^\circ 37' - 34^\circ 29' \\ = 7^\circ 54'$$

$\therefore$ Bearing of B from D
is $97^\circ 54'$

14b)

i) $x = V \cos \theta$

$\therefore t = \frac{x}{V \cos \theta} \quad \text{--- (1)}$

sub in (2) $y = -\frac{1}{2} g t^2 + V t \sin \theta \quad \text{--- (2)}$

$\therefore y = -\frac{g}{2} \times \frac{x^2}{V^2 \cos^2 \theta} + \frac{V \sin \theta \cdot x}{V \cos \theta}$

$\therefore y = \frac{-g}{2V^2 \cos^2 \theta} x^2 + x \tan \theta$

ii)
$$\text{Area} = \int_0^R \left[\frac{-g}{2V^2 \cos^2 \theta} x^2 + x \tan \theta \right] dx$$

$$= \left[\frac{-g x^3}{6V^2 \cos^2 \theta} + \frac{x^2 \tan \theta}{2} \right]_0^R$$

$$= \frac{-g R^3}{6V^2 \cos^2 \theta} + \frac{R^2 \tan \theta}{2}$$

$$= \frac{-g}{3 \cancel{6} V^2 \cos^2 \theta} \times \frac{\cancel{8} V^4 \sin^3 \theta \cos^3 \theta}{g^3 g^2} + \frac{\tan \theta}{\cancel{2}} \times \frac{2^2 V^4 \sin^2 \theta \cos^2 \theta}{g^2}$$

$$= \frac{-4 V^4 \sin^3 \theta \cos \theta}{3 g^2} + \frac{2 V^4 \sin^3 \theta \cos \theta}{g^2}$$

$$= \frac{-4 V^4 \sin^3 \theta \cos \theta + 6 V^4 \sin^3 \theta \cos \theta}{3 g^2}$$

$$\therefore \text{Area} = \frac{2 V^4 \sin^3 \theta \cos \theta}{3 g^2}$$

14 b) iii) $A = \frac{2V^4}{3g^2} \sin^3 \theta \cos \theta$

$$\therefore \frac{dA}{d\theta} = \frac{2V^4}{3g^2} (3 \sin^2 \theta \cos^2 \theta + - \sin^4 \theta) = 0 \text{ st pt}$$

$$\therefore \sin^2 \theta (3 \cos^2 \theta - \sin^2 \theta) = 0$$

$$\therefore 3 \cos^2 \theta = \sin^2 \theta \quad (\theta \text{ is acute})$$

$$\therefore 3 = \tan^2 \theta$$

$$\therefore \tan \theta = \sqrt{3} \quad (\theta \text{ acute})$$

$$\therefore \theta = \frac{\pi}{3}$$

$$(14c) i) \cosh - 1 = 1 - 2 \sin^2 \frac{h}{2} - 1 \\ = -2 \sin^2 \frac{h}{2}$$

$$ii) f(x) = \cos x$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos x \cosh - \sin x \sinh - \cos x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos x (\cosh - 1) - \sin x \sinh}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos x (\cosh - 1)}{h} - \lim_{h \rightarrow 0} \frac{\sin x \sinh}{h}$$

$$= \cos x \lim_{h \rightarrow 0} \frac{\cosh - 1}{h} - \sin x \lim_{h \rightarrow 0} \frac{\sinh}{h}$$

$$= \cos x \lim_{h \rightarrow 0} \frac{-2 \sin^2 \frac{h}{2}}{h} - \sin x \times 1$$

$$= -\cos x \lim_{h \rightarrow 0} \frac{\sin^2 \frac{h}{2}}{\frac{h}{2}} - \sin x$$

$$= -\cos x \lim_{h \rightarrow 0} \frac{\sin \frac{h}{2}}{\frac{h}{2}} \times \frac{\sin \frac{h}{2}}{1} - \sin x$$

$$= -\cos x \times 1 \times 0 - \sin x$$

$$= 0 - \sin x$$

$$= -\sin x$$